

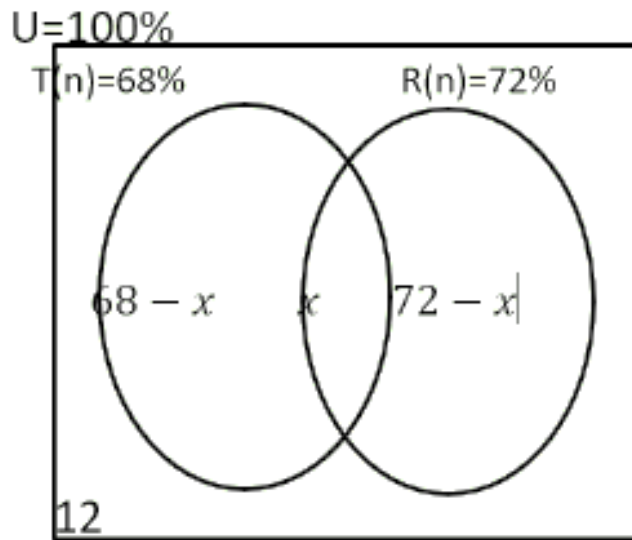
1 (a) i

$U=100\%$ → because percentages are ratio of 100.

$$T(n)=68\%$$

$$R(n)=72\%$$

$$[T(n) \cup R(n)]' = 12\%$$



$U = \text{sum of all region}$

$$\therefore 100 = (68 - x) + (x) + (72 - x) + 12$$

$$100 = 152 - x$$

$$x = 52$$

(ii)

$$\text{television only} = 68 - x$$

$$= 68 - 52$$

$$= 16.$$

(b)

$$Baodu = 4x$$

$$Ansah = 5x$$

change in the ratio = difference in share

$$5x - 4x = 5,000$$

$$x = 5,000$$

$$Baodu = 4x$$

$$= 4(5,000)$$

$$= \mathbf{20,000}.$$

2. (a)

$$p = 2x \sqrt{\frac{q \left(1 + \frac{r^2}{y^2}\right)}{s}}$$

$$\left[p = 2x \sqrt{\frac{q \left(1 + \frac{r^2}{y^2}\right)}{s}} \right]^2$$

$$p^2 = 4x^2 \cdot \frac{q \left(1 + \frac{r^2}{y^2}\right)}{s}$$

$$p^2 s = 4x^2 q \left(1 + \frac{r^2}{y^2}\right)$$

$$\frac{p^2 s}{4x^2} = \frac{4x^2 \cdot q \left(1 + \frac{r^2}{y^2}\right)}{4x^2}$$

$$\left[\frac{p^2 s}{4x^2} = q \left(1 + \frac{r^2}{y^2}\right) \right] \times \frac{1}{q}$$

$$\frac{p^2 s}{4x^2 q} = 1 + \frac{r^2}{y^2}$$

$$\frac{p^2 s}{4x^2 q} = 1 + \frac{r^2}{y^2}$$

$$\left[\frac{p^2 s - 4x^2 q}{4x^2 q} = \frac{r^2}{y^2} \right]^{-1}$$

$$\sqrt{\frac{4x^2 q}{p^2 s - 4x^2 q}} = \sqrt{\frac{y^2}{r^2}}$$

$$2x \sqrt{\frac{q}{p^2 s - 4x^2 q}} = \frac{y}{r}$$

$$\therefore y = 2xr \sqrt{\frac{q}{p^2 s - 4x^2 q}}$$

(b)

$$\begin{aligned} &= \frac{2(3)(-1)^2(-3)}{3(m)-(-2)} \\ &= \frac{-24}{9+2} = \frac{-24}{11} = -2\frac{2}{11} \end{aligned}$$

3. (a)

Note: the sum of angles on a straight line is 180°

$$\therefore 3y + 55 + 2y = 180^\circ$$

$$5y = 180 - 55$$

$$\frac{5y}{5} = \frac{125}{5}$$

$$y = 25^\circ$$

(b)

$$\text{if } \tan x = 1$$

$$x = 45^\circ$$

$$x = 45^\circ$$

$\therefore$

$$\frac{1 - \sin^2 x}{\cos x \cos x}$$

$$\cos^2 x$$

$$1 - \sin^2 x = \cos^2 x \rightarrow \text{Identity}$$

$$\cos^2 x$$

$$\frac{\cos^2 x}{\cos x \cos x} = \cos x$$

$$\cos (45) = 0.707 \text{ or } \frac{\sqrt{2}}{2}$$

4. (a)

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of a pyramid} = \frac{1}{3}(\text{Area of base})h$$

Note: both volumes and height are equal

$$V \text{ of cone} = v \text{ of pyramid}$$

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}(\text{Area of base})h$$

$$\pi r^2 = \text{Area of base}$$

$$\pi r^2 = 154\text{cm}^2$$

$$r^2 = \frac{154^2}{\pi}; \text{ but } \pi = \frac{22}{7}$$

(b)

$$\text{volume} = 6 \text{ liters}$$

$$\text{volume of sphere} = \frac{4}{3}\pi r^3$$

$$\text{quarter volume} = \frac{1}{4}(\text{volume of sphere})$$

$$= \frac{1}{4}\left(\frac{4}{3}\pi r^3\right)$$

$$= \frac{1}{3}\pi r^3.$$

$$\text{quarter volume} = \frac{1}{3}\pi r^3$$

$$6 = \frac{1}{3}\pi r^3$$

$$6 \times 3 = \frac{22}{7} \times r^3$$

$$\frac{18 \times 7}{22} = \frac{22r^3}{22}$$

$$\frac{63}{11} = r^3$$

$$r = \sqrt[3]{\frac{63}{11}} = 0.362$$

5 (a)

$$P = \frac{\text{total number of boys}}{\text{total students}}$$
$$= \frac{32+26+26}{32+26+26+28+44+36}$$

$$= \frac{84}{192} = \frac{7}{16}$$

(b)

$$P = \frac{\text{number of girls in JHS 2}}{\text{total students}}$$
$$= \frac{44}{192}$$
$$= \frac{11}{48}$$

6. (a)

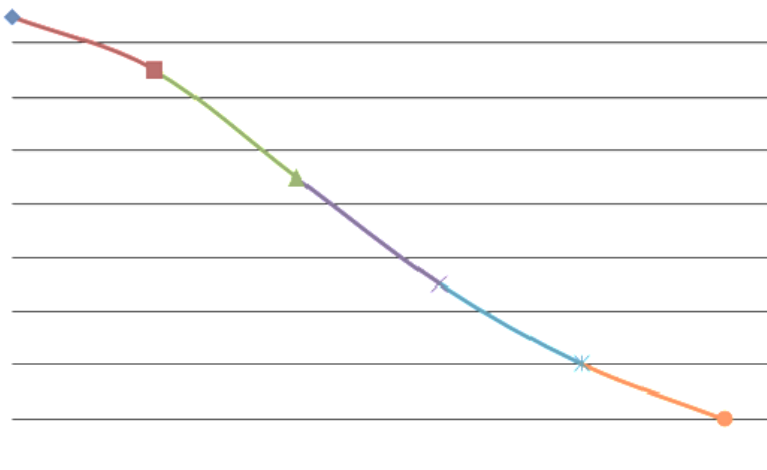
$$a = 7 \cos x - 3 \sin x = 7 \cos (30) - 3(\sin 30) = 4.562 \approx 5$$

$$b = 7 \cos x - 3 \sin x = 7 \cos (60) - 3(\sin 60) = 0.901 \approx 1$$

$$c = 7 \cos x - 3 \sin x = 7 \cos (120) - 3(\sin 120) = -6.09 \approx -6$$

$$d = 7 \cos x - 3 \sin x = 7 \cos (150) - 3(\sin 150) = -6.062 - 1.5 = -7.562 \approx -8$$

(b)



(c)

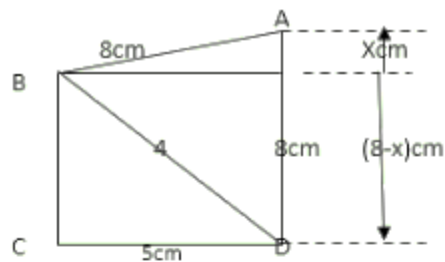
i. $7 \cos x = 3 \sin x$;

this occur at $y=0$, and $x = 67^\circ$

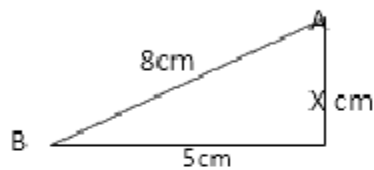
ii. $7 \cos x = 3.2 + 3 \sin x$;

this occur at $y = 3.2$ and $x = 42^\circ$

7. (a)



By Projection



$$c^2 = a^2 + b^2$$

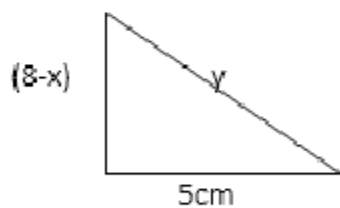
$$8^2 = 5^2 + x^2$$

$$64 - 25 = x^2$$

$$x = \sqrt{39}$$

$$x = 6.2$$

Area of $\triangle BCD$.



$$A = \frac{1}{2} (8 - x)(5)$$

$$A = \frac{1}{2} (8 - 6.2)(5)$$

$$A = 4.5 \text{ cm}^2 \text{ or } 5 \text{ cm}^2$$

$$y = \sqrt{(8 - x)^2 + (5)^2}$$

$$y = \sqrt{(1.8)^2 + (5)^2}$$

$$y = \sqrt{28.54}$$

$$y = 5.3 \text{ or } 5 \text{ cm}$$

(b)

$$\text{Man} = 5(y)$$

$$\text{son} = y$$

In three years: $M = 5y + 3$

$$S = y + 3$$

$$M \times S = 380$$

$$(5y + 3)(y + 3) = 380$$

$$5y^2 + 18y + 9 = 380$$

$$5y^2 + 18y - 371 = 0$$

$$(5y + 53)(y - 7) = 0$$

$$y = -\frac{53}{5} \text{ or } y = 7$$

Hence, we take the positive value

$$\text{son} = y = 7 \text{ yrs}$$

$$\text{Man} = 5y = 5(7) = 35 \text{ yrs.}$$

8. (a)

#of plates = n

$$\text{cost of one plate} = \frac{150}{n}$$

$$\text{number of plates sold} = (n - 4)$$

$$\text{selling price fo one plate} = \frac{180}{n} + 1$$

$$\text{unit profit} = 1$$

$$\text{total profit} = 6$$

$$\text{total amount} = \text{total cost} + \text{total profit}$$

$$= 150 + 6$$

$$= 156$$

$$\text{total amount} = (\text{number of plate sold})(\text{selling profit})$$

$$= (n - 4) \left(\frac{150}{n} + 1 \right)$$

$$\left[156 = (n - 4) \left(\frac{150}{n} + 1 \right) \right] n$$

$$156n = (n - 4)(150 + n)$$

$$156n = 150n + n^2 + 600 - 4n$$

$$n^2 - 10n - 600 = 0$$

$$(n - 30)(n + 20)$$

$$n = 30:$$

(b)

$$\frac{m}{\frac{1}{82}} = \frac{\frac{1}{8}}{m}$$

$$32m = \frac{1}{8m}$$

$$m^2 = \sqrt{\frac{1}{8 \times 32}}$$

$$m = \frac{1}{16}$$

$$\frac{\frac{1}{8}}{m} = \frac{n}{\frac{1}{8}}$$

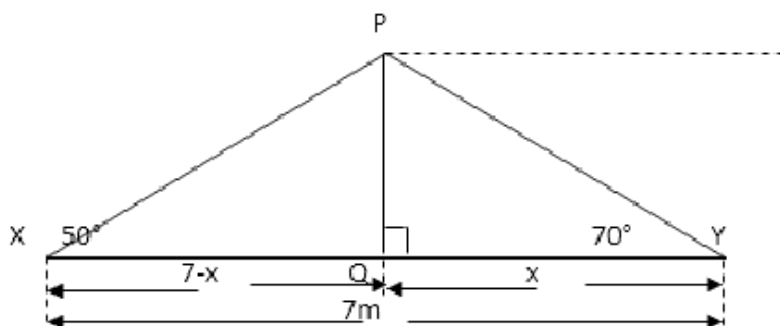
$$8n = \frac{1}{8} \times \frac{1}{m}$$

$$8n = \frac{1}{8} \times \frac{1}{\frac{1}{16}}$$

$$\frac{8n}{8} = \frac{2}{8}$$

$$n = \frac{1}{4}$$

9. (a)



\w

Let $50 = \theta_1$ and $70 = \theta_2$

$$\tan \tan (50) = \frac{n}{(7-x)}$$

$$\text{and } \tan \tan (70) = \frac{n}{x}$$

$$n = (\tan 50)(7-x)$$

$$n = x \tan \tan (70)$$

$$\therefore n = n$$

$$(7-x) \tan \tan 50 = x \tan \tan (70)$$

$$7 \tan \tan 50 = x \tan \tan (70) + x \tan (50)$$

$$8.342 = 3.939x$$

$$x = 2.10 \text{ or } 2$$

$$\text{Then } |XQ| = 7 - x = 7 - 2.1 = 4.9$$

(b)

LCD: of $(1-x)$ and $(1+x)$ is $(1-x^2)$ or $(1-x)(1+x)$

$$\left[\frac{3x}{1-x} + \frac{2x}{x+1} = 2 \right] (1-x)(1+x)$$

$$3x(1+x) + 2x(1-x) = 2(1-x)(1+x)$$

$$3x + 3x^2 + 2x - 2x^2 = 2(1-x^2)$$

$$x^2 + 5x - 2 = 0$$

$$(3x-1)(x+2) = 0$$

$$x = -2 \text{ or } x = \frac{1}{3}$$

10. (b) (i)

$$P = \frac{79}{100}$$

or 0.79

(ii)

$$= 100 - 89$$

$$= 11$$

11. (a)

$$\triangle CDQ = 25^\circ$$

$$25^\circ + 25^\circ + 90^\circ + m = 180^\circ$$

$$140^\circ + m = 180^\circ$$

$$m = 180^\circ - 140^\circ$$

$$m = 40^\circ$$